

Andrea Conti

08 January 1996 - Computer Vision Software Engineer, PhD - Florence, Italy

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Experience

Software Engineer

[Brussels, Belgium](#)

SONY DEPTHSENSING SOLUTIONS

December 2024 - Now

- Design of AI-based computer vision features on cutting-edge camera technologies for the in-vehicle and automotive scenario.
- Multi-sensor integration like IR, RGB and ToF for sensing and automation.

Research Fellowship

[Bologna, Italy](#)

ALMA MATER STUDIORUM

March 2021 - November 2021

- Research grant as part of the *Alma Value - Proof of Concept* program for the enhancement of Alma Mater patents funded by MISE
- Research project focused improving depth measurements from one or more cameras by means of scattered depth data, optionally provided by an active depth sensor. Supervisor: Stefano Mattoccia.

Education

PhD in Computer Science and Engineering

[Bologna, Italy](#)

ALMA MATER STUDIORUM

November 2021 - November 2024

- 3D reconstruction leveraging one or more RGB frames and sparse depth information by means of deep learning approaches in real use-case scenarios. Funded by Sony Depthsensing Solutions.
- Main themes: 3D Sensing, Deep Learning, Sensors Fusion
- Supervisor: Prof. Stefano Mattoccia. Assistant Supervisors: Valerio Cambareri (Sony DepthSensing Solutions NV), Matteo Poggi (UniBo), Paolo Bellavista (UniBo)

Master's Degree in Computer Engineering, 110/110 cum laude

[Bologna, Italy](#)

ALMA MATER STUDIORUM

September 2018 - December 2020

- Final thesis project "*Diving between depth prediction and depth completion*" focused on the application of deep neural networks to the monocular perception of depth with the optional support of lidar sensors.
- Main themes: Deep Learning, LIDAR sensors, Depth Prediction, Depth Completion
- Supervisor: Prof. Stefano Mattoccia. Assistant supervisors: Matteo Poggi, Filippo Aleotti, Fabio Tosi

Bachelor Degree in Computer Engineering, 110/110 cum laude

[Bologna, Italy](#)

ALMA MATER STUDIORUM

September 2015 - October 2018

- Final thesis project "*Confidence Measurements for Embedded Systems relying on Machine Learning*" focused on the application of artificial intelligence techniques to the prediction of the confidence of disparity maps taking into account efficiency.
- Main themes: Machine Learning, Depth Prediction, Decision Trees
- Supervisor: Prof. Stefano Mattoccia. Assistant supervisors: Matteo Poggi, Fabio Tosi

Research Activity

RESEARCH TOPICS

The primary research focus is on **3D reconstruction** utilizing deep learning and machine learning techniques with various input sources in challenging real-world environments. This encompasses extensive expertise in **stereo vision**, **multi-view stereo**, **sensor fusion** with **active sensors**, and **optical flow**.

PUBLICATIONS

ToF-Splatting: Dense SLAM using Sparse Time-of-Flight Depth and Multi-Frame Integration

[Honolulu, Hawaii](#)

INTERNATIONAL CONFERENCE ON COMPUTER VISION (ICCV)

October 2025

A. Conti, M. Poggi, V. Cambareri, Martin R. Oswald, S. Mattoccia

Depth on Demand: Streaming Dense Depth from a Low Frame Rate Active Sensor

[Milan, Italy](#)

EUROPEAN CONFERENCE ON COMPUTER VISION (ECCV)

October 2024

A. Conti, M. Poggi, V. Cambareri, S. Mattoccia

LiDAR-Event Stereo Fusion with Hallucinations

[Milan, Italy](#)

EUROPEAN CONFERENCE ON COMPUTER VISION (ECCV)

October 2024

L. Bartolomei, M. Poggi, A. Conti, S. Mattoccia

Range-Agnostic Multi-View Depth Estimation With Keyframe Selection

INTERNATIONAL CONFERENCE ON 3D VISION (3DV)
A. Conti, M. Poggi, V. Cambareri, S. Mattoccia

Davos, Switzerland

March 2024

Revisiting Depth Completion from a Stereo Matching Perspective for Cross-Domain Generalization

INTERNATIONAL CONFERENCE ON 3D VISION (3DV)
L. Bartolomei, M. Poggi, A. Conti, F. Tosi, S. Mattoccia

Davos, Switzerland

March 2024

Active Stereo Without Pattern Projector

IEEE/CVF INTERNATIONAL CONFERENCE ON COMPUTER VISION (ICCV)
L. Bartolomei, M. Poggi, F. Tosi, A. Conti, S. Mattoccia

Paris, France

October 2023

Boosting Multi-Modal Unsupervised Domain Adaptation for LiDAR Semantic Segmentation by Self-Supervised Depth Completion

IEEE ACCESS, VOL. 11, PP. 85155-85164
A. Cardace, A. Conti, P. Z. Ramirez, R. Spezialetti, S. Salti and L. D. Stefano

Journal

August 2023

Sparsity Agnostic Depth Completion

IEEE/CVF WINTER CONFERENCE ON APPLICATIONS OF COMPUTER VISION (WACV)
A. Conti, M. Poggi, S. Mattoccia

Waikoloa, Hawaii

January 2023

Unsupervised confidence for LiDAR depth maps and applications

IEEE/RSJ INTERNATIONAL CONFERENCE ON INTELLIGENT ROBOTS AND SYSTEMS (IROS)
A. Conti, M. Poggi, F. Aleotti, S. Mattoccia

Kyoto, Japan

October 2022

Multi-View Guided Multi-View Stereo

IEEE/RSJ INTERNATIONAL CONFERENCE ON INTELLIGENT ROBOTS AND SYSTEMS (IROS)
M. Poggi*, A. Conti*, S. Mattoccia. *joint authorship

Kyoto, Japan

October 2022

Monitoring social distancing with single image depth estimation

IEEE TRANSACTIONS ON EMERGING TOPICS IN COMPUTATIONAL INTELLIGENCE (TETCI)
A. Mingozzi, A. Conti, F. Aleotti, M. Poggi, S. mattoccia

Journal

April 2022

On Deployment of Out-of-the-Box Embedded Devices for Self-Powered River Surface Flow Velocity Monitoring at the Edge

MDPI APPLIED SCIENCE
A. H. Livoroi, A. Conti, L. Foianesi, F. Tosi, F. Aleotti, M. Poggi, F. Tauro, E. Toth, S. Grimaldi and S. Mattoccia

Journal

May 2021

PARTECIPATION IN CONFERENCES

Prizes

OUTSTANDING REVIEWER AT CVPR

2024

Acknowledged as being among the top 2% of reviewers, as evaluated by the Area Chairs, out of a total of 9,872 reviewers. This recognition was awarded in appreciation of the high-quality and insightful reviews provided, demonstrating a commitment to academic rigor and contributing to the advancement of knowledge within the field.

Presentations

EUROPAN CONFERENCE ON COMPUTER VISION (ECCV)
INTERNATIONAL CONFERENCE ON 3D VISION (3DV)
IEEE/CVF INTERNATIONAL CONFERENCE ON COMPUTER VISION (ICCV)
IEEE/CVF WINTER CONFERENCE ON APPLICATIONS OF COMPUTER VISION (WACV)
IEEE/RSJ INTERNATIONAL CONFERENCE ON INTELLIGENCE ROBOTS AND SYSTEMS (IROS)

Milan, Italy, 2024
Davos, Switzerland, 2024
Paris, France, 2023
Waikoloa, Hawaii, 2023
Kyoto, Japan, 2022

Reviewing Service

IEEE/CVF INTERNATIONAL CONFERENCE ON COMPUTER VISION (ICCV)
IEEE/CVF CONFERENCE ON COMPUTER VISION AND PATTERN RECOGNITION (CVPR)
IEEE/RSJ INTERNATIONAL CONFERENCE ON INTELLIGENCE ROBOTS AND SYSTEMS (IROS)
EUROPEAN CONFERENCE ON COMPUTER VISION (ECCV)

2023
2022 - 2023 - 2024 - 2025
2022 - 2023
2022 - 2024

Skills and Background Knowledge

COMPUTER VISION & DEEP LEARNING TOOLS

- Advanced knowledge of multi-view geometry and related tasks like stereo vision, multi-view stereo and optical flow, as well as common issues and common solutions
- Good knowledge of the mainstream tools for deep learning development: Pytorch, Pytorch Lightning, Tensorflow, Keras, JAX
- Other tools and technologies for visualization and machine learning other than deep learning in Python NumPy, SciPy, Pandas, Scikit-Learn, Seaborn, Matplotlib, Scikit-Image, MIFlow, WanDB, Numba

DEVOPS & SYSTEMS ADMINISTRATION

- Good knowledge unix system administration tools, scripting languages such as *Bash* and *Fish*, tools like *ssh*, *tmux*, *openvpn*, *iptables*
- Good knowledge of virtualization tools such as virtual machines and *Docker*
- Excellent knowledge of *Git*

SOFTWARE ENGINEERING

- Deep mastery of Python programming, concepts and underlying mechanisms
- Knowledge of various programming paradigms studied in a heterogeneous set of programming languages: imperative programming, object-oriented programming (Python, Java, C), functional programming (Haskell, Elixir, Clojure) and message-passing programming (Elixir and Golang).

LANGUAGES

- **Italiano** native language.
- **English** fluent writing and reading, good speaking skills (B2 certificate).

Other Activities

SCHOOLS ATTENDED

Deep Learning and Computer Vision School

FRANCESCA ODONE, NOCETI NICOLETTA

Genova, Italy

July 2023

Advanced Methods for Mathematical Image Analysis

LUCA CALTRONI (CNRS), JACEK GONDZIO (U. EDINBURG), OZAN OKTEM (KTH), SAMULI SILTATEN (U. HELSINKI)

Bologna, Italy

January 2023

Bertinoro International Spring School

MARCO GORI (U. SIENA), ARISTIDES GIONIS (KTH), MASSIMO VILLARI (U. MESSINA)

Bertinoro, Italy

March 2022

TUTOR ACTIVITIES

Electronic Calculators

ALMA MATER STUDIORUM

Bologna, Italy

September 2021 - September 2024

Fundamentals of Computer Science

ALMA MATER STUDIORUM

Bologna, Italy

February 2021 - September 2021

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